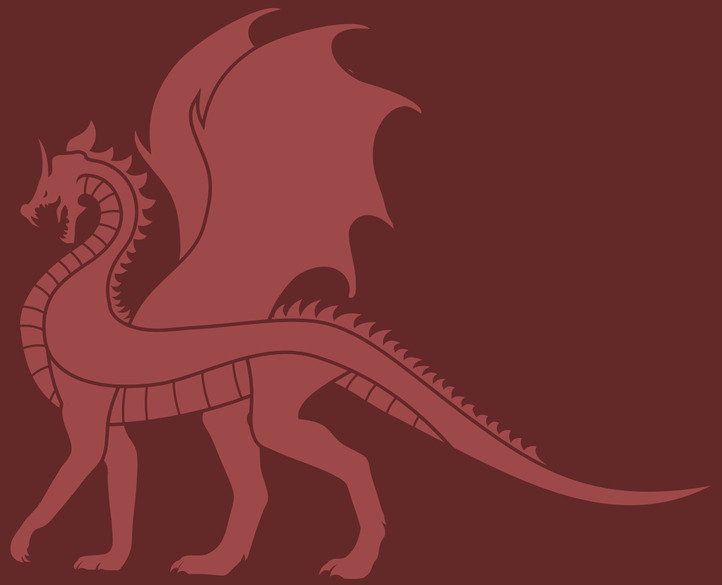
Dissertation

Application of the Internet of Things in Game Development and Interactive Toys: A Comprehensive Review and Future Directions



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2022BALA07ALA006

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Content

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Key Words

- Internet of Things
- Pervasive Games
- Serious Games
- Phygital Objects
- Simulation



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Research Objective

- To investigate the various ways in which IoT is used in game development and interactive toys.
- To explore the impact of games developed using IoT.
- To use IoT to design a new form of Gameplay.

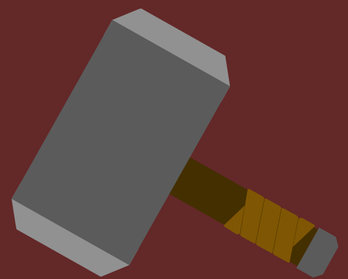


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Methodology

- The research project is divided into three parts within two chapters.
- In the findings chapter, the first two parts describes how IoT is used to develop games and interactive toys, and explores the impact of the products mention in the other work on their consumers.
- The discussion chapter consists of the third part which tries to define a new way of using IoT in Game development.

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IoT in Game Development and Interactive Toys



Network Architecture for Game Development



Network Architecture for Interactive Toys

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The Impact of IoT in the Field

Game Development

- Serious games used for medical education have proved to increase the cognitive skills of medical students within an engaging environment (de Ribaupierre et al, 2014).
- The immersive experience of pervasive Game makes the process less tedious and increases engagement, enabling the patient to recover completely (Arango-López et al., 2019).

Interactive Toys

- LOVOTs communicate through non-verbal communication and body language, they are highly inclusive for older adults who are socially isolated or have cognitive impairments (Hung et al., 2025).
- Cozmo is capable of instilling curiosity, creativity, and a competitive nature within children (Kay et al, 2023).

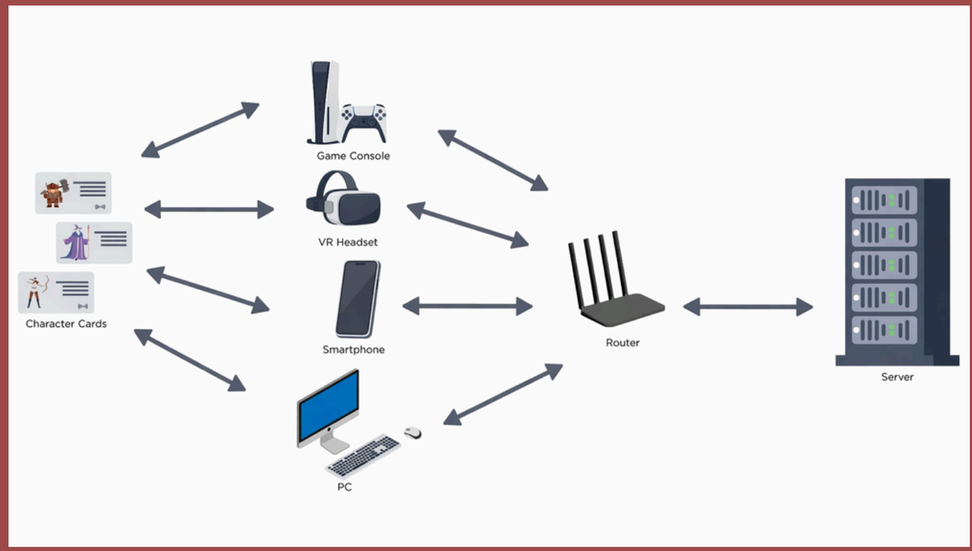
Another way of using IoT in Game Development

Devices in the network

1. Character card
2. Console
3. PC
4. Smartphone
5. VR Headset
6. Router
7. Server

Network Architecture

- The character cards are used to access the player's avatar using NFC.
- The character card is used to authenticate the use of the Avatar.
- The console, VR headset, Smartphone and PC use NFC to grant the player the use of the Avatar on the character card
- The router is used to communicate the player's interaction to the server.
- The server processes the player's interaction and reponds with the players desired action in realtime.



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Another way of using IoT in Game Development

Smartphone

- The Smartphone Narrative is similar to Pokemon GO.
- The player can roam around in the real world collecting raw materials from ores and trees.
- They can also loot chest for rare items that can not be crafted.
- The players have a limited amount of stamina which they use collect raw material and open chests.
- The Map includes territories of hostile Non-Playable Characters who try to steal the players' collected raw material.
- If a player does stumble into a hostile NPC, They can use the escape option to evade them at the cost of their stamina. If the player fails to escape the NPC, the NPC will steal a random portion of their raw material.
- The Player can also choose to fight the hostile NPC with a weapon they found in a chest or their skill from the Console and PC narrative.



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Another way of using IoT in Game Development

PC and Console

- In the PC and Console Narrative, the players are in an open world map, where the players can roam freely.
- The main Objective for the players is to complete quests.
- Players can explore various biomes and interact with friendly and hostile Non-Playable Character (NPC).







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Another way of using IoT in Game Development

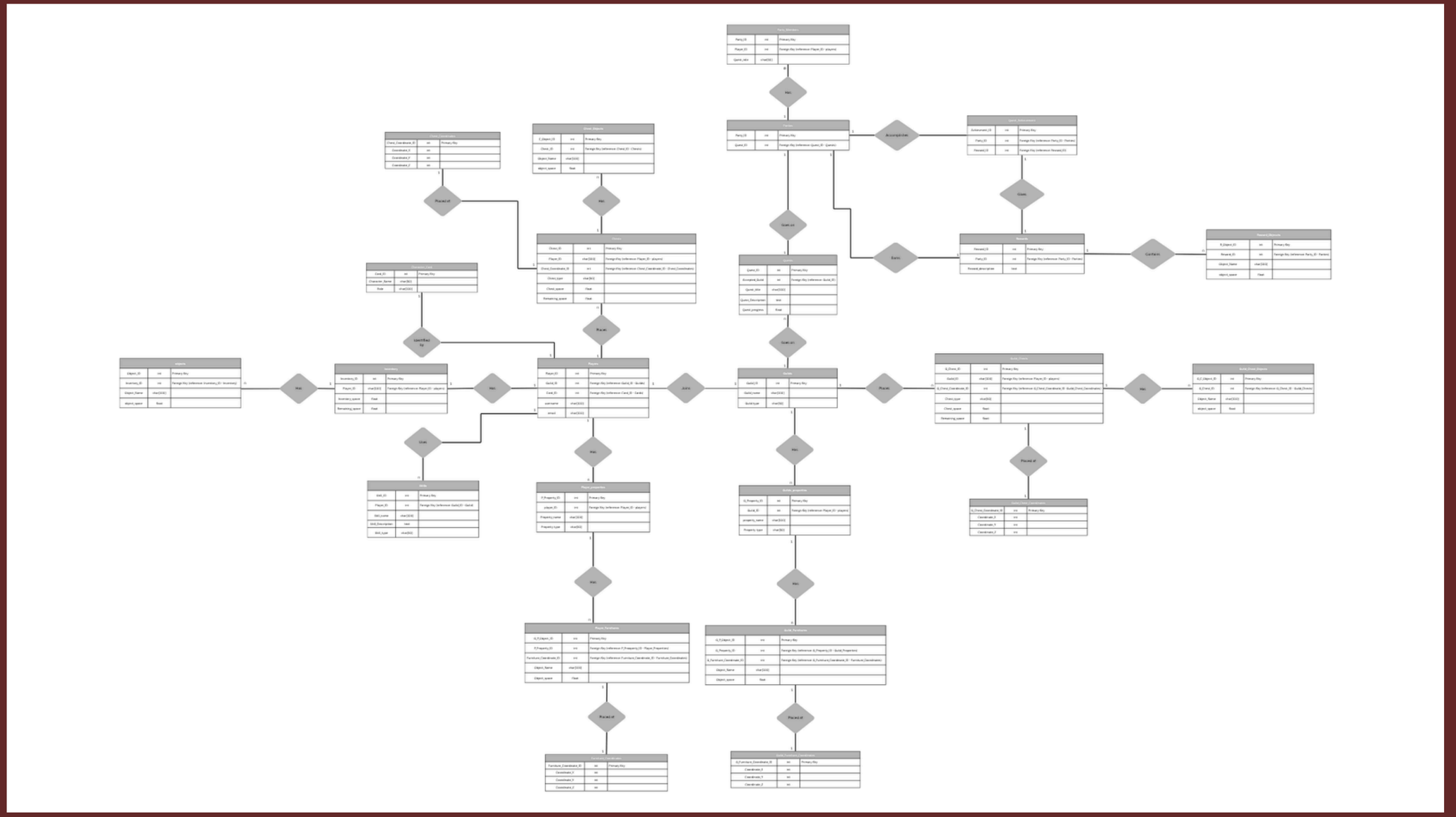
Virtual Reality Headset

- In the VR Narrative of the game, the players can build Guildhalls, ships, and other building in player built villages.
- Players can place chests, tables, chairs and other furniture within the building they are building.
- The buildings that players build in the VR narrative are translated to specific coordinated where players can free.



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ER Diagram



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Entity Tables - Players

Character_Card		
Card_ID	int	Primary Key
Character_Name	char[50]	
Role	char[100]	

Players		
Player_ID	int	Primary Key
Guild_ID	int	Foreign Key (reference: Guild_ID - Guilds)
Card_ID	int	Foreign Key (reference: Card_ID - Cards)
username	char[100]	
email	char[100]	

Inventory		
Inventory_ID	int	Primary Key
Player_ID	char[100]	Foreign Key (reference: Player_ID - players)
Inventory_space	float	
Remaining_space	float	

objects		
Object_ID	int	Primary Key
Invertory_ID	int	Foreign Key (reference: Inventory_ID - Inventory)
Object_Name	char[100]	
object_space	float	

Skills		
Skill_ID	int	Primary Key
Player_ID	int	Foreign Key (reference: Player_ID - Players)
Skill_name	char[100]	
Skill_Description	text	
Skill_type	char[50]	

Player_Furnitures		
P_P_Object_ID	int	Primary Key
P_Property_ID	int	Foreign Key (reference: P_Prooperty_ID - Player_Properties)
P_P_Furniture_Coordinate_ID	int	Foreign Key (reference: P_P_Furniture_Coordinate_ID - Furniture_Coordinates)
Object_Name	char[100]	
Object_space	float	

Furniture_Coordinates		
P_P_Furniture_Coordinate_ID	int	Primary Key
Coordinate_X	int	
Coordinate_Y	int	
Coordinate_Z	int	

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Entity Tables - Player Chests and properties

Chests		
Chest_ID	int	Primary Key
Player_ID	char[100]	Foreign Key (reference: Player_ID - players)
Chest_Coordinate_ID	int	Foreign Key (reference: Chest_Coordinate_ID - Chest_Coordinates)
Chest_type	char[50]	
Chest_space	float	
Remaining_space	float	

Player_properties		
P_Property_ID	int	Primary Key
player_ID	int	Foreign Key (reference: Player_ID - players)
P_Property_Coordinate_ID	int	Foreign Key (reference: P_Property_Coordinate_ID - P_Property_Coordinates)
Property_name	char[100]	
Property type	char[50]	

Chest_Coordinates		
Chest_Coordinate_ID	int	Primary Key
Coordinate_X	int	
Coordinate_Y	int	
Coordinate_Z	int	

P_Property_Coordinates		
P_Property_Coordinate_ID	int	Primary Key
Coordinate_X	int	
Coordinate_Y	int	
Coordinate_Z	int	

Chest_Objects		
C_Object_ID	int	Primary Key
Chest_ID	int	Foreign Key (reference: Chest_ID - Chests)
Object_Name	char[100]	
object_space	float	

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Entity Tables - Guilds

Guilds		
Guild_ID	int	Primary Key
Guild_name	char[100]	
Guild type	char[50]	

Guild_Furnitures		
G_P_Object_ID	int	Primary Key
G_Property_ID	int	Foreign Key (reference: G_Property_ID - Guild_Properties)
G_P_Furniture_Coordinate_ID	int	Foreign Key (reference: G_P_Furniture_Coordinate_ID - Furniture_Coordinates)
Object_Name	char[100]	
Object_space	float	

Guild_Chests		
G_Chest_ID	int	Primary Key
Guild_ID	char[100]	Foreign Key (reference: Player_ID - players)
G_Chest_Coordinate_ID	int	Foreign Key (reference: G_Chest_Coordinate_ID -Guild_Chest_Coordinates)
Chest_type	char[50]	
Chest_space	float	
Remaining_space	float	

Guild_Furniture_Coordinates		
G_P_Furniture_Coordinate_ID	int	Primary Key
Coordinate_X	int	
Coordinate_Y	int	
Coordinate_Z	int	

Guilds_properties		
G_Property_ID	int	Primary Key
Guild_ID	int	Foreign Key (reference: Player_ID - players)
G_P_Property_Coordinate_ID	int	Foreign Key (reference: G_P_Property_Coordinate_ID - G_Property_Coordinates)
property_name	char[100]	
Property type	char[50]	

Guild_Chest_Objects		
G_C_Object_ID	int	Primary Key
G_Chest_ID	int	Foreign Key (reference: G_Chest_ID - Guild_Chests)
Object_Name	char[100]	
object_space	float	

Guild_Chest_Coordinates		
G_Chest_Coordinate_ID	int	Primary Key
Coordinate_X	int	
Coordinate_Y	int	
Coordinate_Z	int	

G_Property_Coordinates		
G_Property_Coordinate_ID	int	Primary Key
Coordinate_X	int	
Coordinate_Y	int	
Coordinate_Z	int	

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Entity Tables - Quests

Quests		
Quest_ID	int	Primary Key
Accepted_Guild	int	Foreign Key (reference: Guild_ID)
Quest_title	char[100]	
Quest_Description	text	
Quest_progress	float	

Quest_Achievement		
Achievment_ID	int	Primary Key
Party_ID	int	Foreign Key (reference: Party_ID - Parties)
Reward_ID	int	Foreign Key (reference: Reward_ID)

Rewards		
Reward_ID	int	Primary Key
Reward_description	text	

Parties		
Party_ID	int	Primary Key
Quest_ID	int	Foreign Key (reference: Quest_ID - Quests)
Party_Member_ID	int	Foreign Key (reference: Quest_ID - Quests)

Party_Members		
Party_Member_ID	int	Primary Key
Player_ID	int	Foreign Key (reference: Player_ID - players)
Party__ID	int	Foreign Key (reference: Party__ID - Parties)
Quest_role	char[50]	

Reward_Objeccts		
R_Object_ID	int	Primary Key
Reward_ID	int	Foreign Key (reference: Party_ID - Parties)
Object_Name	char[100]	
object_space	float	

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